

• HYBRID PAY IMPLEMENTATION STEPS

Prerequisites

- Java 17 or later version installed
- Git
- MySQL locally or Railway DB

STEPS:

1) Get the code

```
git clone https://github.com/Keerthana2k24/HybridPay--A_Digital_Payment_Processing_Platform.git
cd HybridPay--A_Digital_Payment_Processing_Platform
```

2) Configure Database

Option A - Local MySQL

Create DB: **CREATE DATABASE hybridpay;**

Then:

Update application.properties (LOCAL RUN)

- Use this when running on your system.

```
server.port=8080
spring.datasource.url=jdbc:mysql://localhost:3306/hybridpay?useSSL=false&allowPublicKeyRetrieval=true
spring.datasource.username=root
spring.datasource.password= your_password
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
spring.jpa.properties.hibernate.format_sql=true
spring.jpa.database-platform=org.hibernate.dialect.MySQLDialect
```

3) Run Backend (Spring Boot)

./mvnw spring-boot:run

4) Open the App

Go to: **http://localhost:8080**

Option B - Use Railway (easiest)

Take values from Railway Variables:

- MYSQLHOST

- MYSQLPORT
- MYSQLDATABASE
- MYSQLUSER
- MYSQLPASSWORD

Update src/main/resources/application.properties:

- Use this for deployment (secure way).

server.port=8080

spring.datasource.url= // URL from deployed site (Railway Mysql)

spring.datasource.username=root

spring.datasource.password= // Your Password

spring.jpa.hibernate.ddl-auto= update

spring.jpa.show-sql=true

spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQLDialect

spring.jpa.properties.hibernate.format_sql=true

5) Test flow

1. Register two accounts (User A and User B)
2. User A gets ₹10,000 welcome bonus
3. Log in as User A → Send Money → enter User B's email → send ₹500
4. Log out → Log in as User B → see ₹500 received
5. Both users see the transaction in History

• HYBRID PAY DEPLOYMENT STEPS

1. Railway (MySQL Database Setup)

◇ Step 1: Create DB

- Go to [Railway.com](https://railway.com) -> LOGIN-> **New Project**
- Click → **Provision MySQL**

◇ Step 2: Get credentials

Go to: 🖱️ **Variables tab**

You'll see:

MYSQLHOST
MYSQLPORT
MYSQLDATABASE
MYSQLUSER
MYSQLPASSWORD

◇ **Step 3: Keep these ready (update in application.properties)**

DB_URL=jdbc:mysql://...
DB_USER=root
DB_PASS=your_password

2. Render (Spring Boot Backend)

◇ **Step 1: Push code to GitHub**

```
git add .  
git commit -m "deploy"  
git push
```

◇ **Step 2: Create service in Render**

- Go to [Render.com](https://render.com) → **New** → **Web Service**
- Connect your GitHub repo

◇ **Step 3: Choose environment**

Select: 🖱️ **Docker**

◇ **Step 4: Add Dockerfile (important)**

```
FROM eclipse-temurin:17-jdk
```

```
WORKDIR /app  
COPY . .
```

```
RUN chmod +x mvnw  
RUN ./mvnw clean package -DskipTests
```

```
CMD ["java", "-jar", "target/hybridpay-1.0.0.jar"]
```

◇ **Step 5: Add Environment Variables**

In Render → **Environment**

DB_URL=jdbc:mysql://tramway.proxy.rlwy.net:42149/railway?useSSL=false&allowPublicKeyRetrieval=true

DB_USER=root

DB_PASS=your_password

[Copy from application.properties file and paste it in Environment variable]

◇ Step 7: Deploy

Click: 👉 **Create Web Service**

Render will:

- Build Docker image
- Run Spring Boot
- Give public URL

3. Final Check

Open: <https://your-app.onrender.com>